

Research Interests

My research lies at the intersection of computer graphics and vision, with prior work spanning human digitization, reflectance capture, and material appearance modeling. Recently, I have focused on **human-centric visual computing**, developing representations that combine physical principles with data-driven learning to capture human geometry, appearance, motion, and interactions with the physical world. My long-term goal is to enable high-fidelity, controllable, and physically grounded visual representations that support embodied AI, next-generation telepresence, and high-end visual media.

Education

- 2019 – 2025 **Yale University**, New Haven, USA
Ph.D. in Computer Science
Advisor: **Prof. Holly Rushmeier**
Dissertation: *Neural Representations for Digital Humans*
- 2015 – 2019 **Zhejiang University**, Hangzhou, China
B.E. in Computer Science and Technology
Advisor: **Prof. Hongzhi Wu**
Thesis: *Image-based Reconstruction of High-precision 3D Geometry and Material*

Employment

- 2025 – **Meta**, Pittsburgh, USA
Research Scientist
Topic: Relighting
- Summer 2024 **Meta**, Pittsburgh, USA
Research Scientist Intern advised by **Dr. Giljoo Nam**
Topic: Universal 3D hair modeling
- Summer 2023 **Tencent AI Lab**, Shenzhen, China
Research Intern advised by **Dr. Di Kang**
Topic: Controllable 3D avatar synthesis
- Summer 2022 **Adobe Research**, San Jose, USA
Research Scientist Intern advised by **Dr. Yi Zhou**
Topic: Parametric model of 3D hair
- Summer 2021 **Adobe Research**, San Jose, USA
Research Intern advised by **Dr. Yi Zhou** and **Dr. Jun Saito**
Topic: Neural motion fields
- Summer 2019 **miHoYo**, Shanghai, China
Research Intern advised by **Dr. Jun Xing**
Topic: Stylized texture synthesis

Publications

- 1. Large-scale Codec Avatars: The Unreasonable Effectiveness of Large-scale Avatar Pretraining**
Meta Codec Avatars Lab
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

2. **CamLit: Unified Video Diffusion with Explicit Camera and Lighting Control**
Zhiyi Kuang, **Chengan He**, Egor Zakharov, Yuxuan Xue, Shunsuke Saito, Olivier Maury, Timur Bagautdinov, Youyi Zheng, and Giljoo Nam
arXiv, 2026
3. **Digital Salon: An AI and Physics-Driven Tool for 3D Hair Grooming and Simulation**
Chengan He*, Jorge Alejandro Amador Herrera*, Zhixin Shu, Xin Sun, Yao Feng, Sören Pirk, Dominik L. Michels, Meng Zhang, Tuanfeng Y. Wang, Julie Dorsey, Holly Rushmeier, Yi Zhou
arXiv, 2025
4. **Augmented Mass-Spring Model for Real-Time Dense Hair Simulation**
Jorge Alejandro Amador Herrera, Yi Zhou, Xin Sun, Zhixin Shu, **Chengan He**, Sören Pirk, Dominik L. Michels
IEEE/CVF International Conference on Computer Vision (ICCV), 2025
5. **3DGH: 3D Head Generation with Composable Hair and Face**
Chengan He, Junxuan Li, Tobias Kirschstein, Artem Sevastopolsky, Shunsuke Saito, Qingyang Tan, Javier Romero, Chen Cao, Holly Rushmeier, Giljoo Nam
ACM Transactions on Graphics (SIGGRAPH), 2025
6. **PERM: A Parametric Representation for Multi-Style 3D Hair Modeling**
Chengan He, Xin Sun, Zhixin Shu, Fujun Luan, Sören Pirk, Jorge Alejandro Amador Herrera, Dominik L. Michels, Tuanfeng Y. Wang, Meng Zhang, Holly Rushmeier, Yi Zhou
International Conference on Learning Representations (ICLR), 2025 (Spotlight)
7. **Chang-E: A High-Quality Motion Capture Dataset of Chinese Classical Dunhuang Dance**
Zeyu Wang, **Chengan He**, Zhe Yan, Jiashun Wang, Yingke Wang, Junhua Liu, Angela Shen, Mengying Zeng, Holly Rushmeier, Huazhe Xu, Borou Yu, Chenchen Lu, Eugene Y. Wang
ACM Journal on Computing and Cultural Heritage (JOCCH), 2025
8. **NeMF: Neural Motion Fields for Kinematic Animation**
Chengan He, Jun Saito, James Zachary, Holly Rushmeier, Yi Zhou
Advances in Neural Information Processing Systems (NeurIPS), 2022 (Spotlight)
9. **An Inverse Procedural Modeling Pipeline for SVBRDF Maps**
Yiwei Hu, **Chengan He**, Valentin Deschaintre, Julie Dorsey, Holly Rushmeier
ACM Transactions on Graphics (SIGGRAPH), 2022
10. **Learning Efficient Illumination Multiplexing for Joint Capture of Reflectance and Shape**
Kaizhang Kang, Cihui Xie, **Chengan He**, Mingqi Yi, Minyi Gu, Zimin Chen, Kun Zhou, Hongzhi Wu
ACM Transactions on Graphics (SIGGRAPH Asia), 2019

Live Demos

1. **Digital Salon: An AI and Physics-Driven Tool for 3D Hair Grooming and Simulation**
Chengan He*, Jorge Alejandro Amador Herrera*, Yi Zhou*, Zhixin Shu, Xin Sun, Yao Feng, Sören Pirk, Dominik L. Michels, Meng Zhang, Tuanfeng Y. Wang, Holly Rushmeier
SIGGRAPH Asia 2024 Real-Time Live!
2. **Project Motion Mix: Human Animation in a Loop from Images**
Jae Shin Yoon, Duygu Ceylan, Tuanfeng Y. Wang, Jingwan Lu, Jimei Yang, Zhixin Shu, **Chengan He**, Yi Zhou, Jun Saito, James Zachary
Adobe Max Sneaks 2022

Patents

1. **Parametric Hair Model for Digital Hair Generation**
US Patent App. 18/769233
2. **Generating Three-Dimensional Looping Animations from Still Images**
US Patent App. 18/340445

Invited Talks

Teaching

Spring 2024	CPSC 482/582: Current Topics in Applied Machine Learning	Teaching Assistant
Fall 2023	CPSC 446/546: Data and Information Visualization	Teaching Assistant
Spring 2023	CPSC 480/580: Introduction to Computer Vision	Teaching Assistant
Spring 2022	CPSC 276/376: Digital Humanities	Teaching Assistant
Fall 2021	CPSC 446/546: Data and Information Visualization	Teaching Assistant

Professional Services

Reviewer	ACM SIGGRAPH, ACM SIGGRAPH Asia, Eurographics, Pacific Graphics, CAD/Graphics, CVPR, ICCV, ECCV, WACV, NeurIPS, ICLR, ICML, AAAI, The Visual Computer, IEEE Transactions on Visualization and Computer Graphics
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Selected Awards and Honors

2025	Doctoral Consortium, ICCV
2025	Outstanding Reviewer, CVPR
2018	National Scholarship, China Ministry of Education